STAT 436 HW1

Solutions

March 2023

```
library(tidyverse)
library(lubridate)
theme_set(theme_bw())
```

Reshaping Practice [3 points]

The following parts ask you to provide R code that transforms one dataset (I) into another (II).

a. Regional murder rates: Turn I into II.

We can use the code below. The main observation is that we need aggregate across all states within a region, and this is a natural fit for the `group_by` + `summarise` pattern. The murder rate within a region is defined as the total number of murders over the total population within that region. Since we don't have these totals defined in advance, we first compute them using the first two lines within the `summarise` call. Once those two calls, the new columns are available to define `murder_rate` in the third line. Finally, we notice that the example data are sorted in order from lowest to highest murder rates. To ensure this ordering, we can use `arrange`.

Note that, in general, any transformations defined at the start of a `summarise` or `mutate` call can be used in subsequent arguments.

```
read_csv("https://uwmadison.box.com/shared/static/h4gau9heqy3uue9rmpq9b55s3kw37zd8.csv") %>%
  group_by(region) %>%
  summarise(
    murders = sum(total),
    population = sum(population),
    murder_rate = murders / population
  ) %>%
  arrange(murder_rate)
```

```
## # A tibble: 4 x 4
##   region        murders population murder_rate
##   <chr>           <dbl>      <dbl>       <dbl>
## 1 Northeast        1469   55317240   0.0000266
## 2 West             1911   71945553   0.0000266
## 3 North Central    1828   66927001   0.0000273
## 4 South            4195  115674434   0.0000363
```

b. Antibiotics and bacteria. Turn I into II

We can use the code below. Our first observation is that, while the the different species names are spread out across columns in dataset (I), they are stacked as different rows in (I). We can reshape the data accordingly

using `pivot_longer()`. The `starts_with("Unc")` selector ensures that every column that stored a species name (which all start with `"Unc"`) gets merged into a unified, stacked column.

Our next observation is that the sample names have been split into subject name and sampling times. There are several ways we could introduce this information – e.g., processing the string names within `mutate()` – but the most concise approach is to call `separate` with the third argument used to denote the character position at which to split the sample names. Finally, as in part (b), we use `arrange()` to sort the rows into the order seen in (II).

```
read_csv("https://uwmadison.box.com/shared/static/bq4afq9kl2zn9qlb89q2rxhrrv73iuil.csv") %>%
  pivot_longer(starts_with("Unc"), names_to = "species") %>%
  separate(sample, c("ind", "time"), 1, convert = TRUE) %>%
  arrange(ind, species, time)
```

```
## # A tibble: 972 x 5
##    ind    time antibiotic      species  value
##    <chr> <int> <chr>           <chr>    <dbl>
##  1 D         1 Antibiotic-free Unc05qi6     0
##  2 D         2 Antibiotic-free Unc05qi6     0
##  3 D         3 Antibiotic-free Unc05qi6     0
##  4 D         4 Antibiotic-free Unc05qi6     0
##  5 D         5 Antibiotic-free Unc05qi6     0
##  6 D         6 Antibiotic-free Unc05qi6     0
##  7 D         7 Antibiotic-free Unc05qi6     0
##  8 D         8 Antibiotic-free Unc05qi6     1
##  9 D         9 Antibiotic-free Unc05qi6     0
## 10 D        10 Antibiotic-free Unc05qi6     0
## # ... with 962 more rows
```

City Temperatures [3 points]

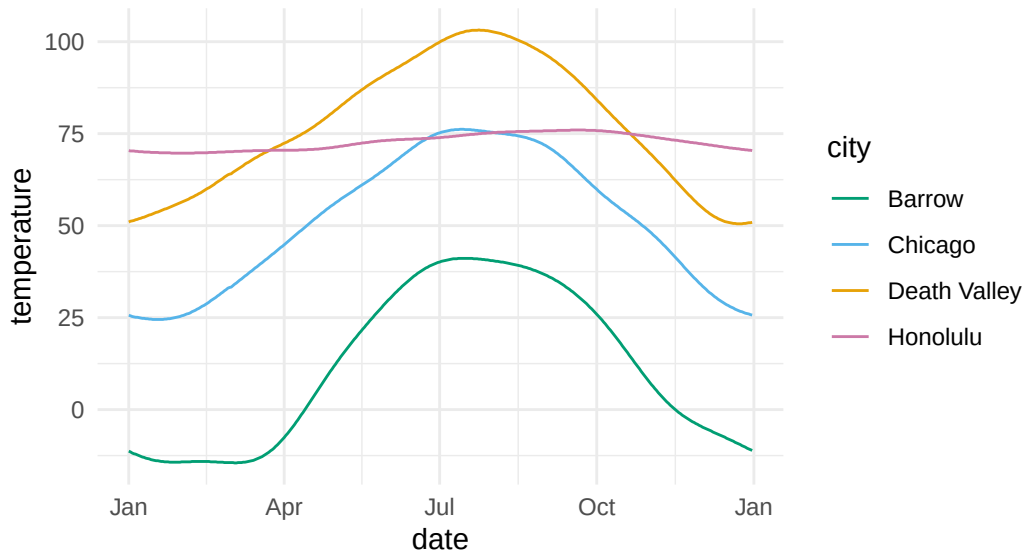
Let's create versions of Figure 2.3 and 2.4 from the reading this week. The command below reads in the data. We've filtered to a slightly different set of cities (Barrow is in Alaska, Honolulu is in Hawaii), but we should still be able to study changes in temperature over time.

```
temperature <- read_csv("https://raw.githubusercontent.com/krisrs1128/stat479_s22/main/data/temperature
```

a. Make a version of Figure 2.3 using a line mark (`geom_line`). Make at least one customization of the theme to make the plot more similar to the version in Figure 2.3. Hint: To group the lines by city, use the `group =` aesthetic mapping.

This is a line plot with with the `x`, `y`, and `color` channels encoding time, temperature, and city, respectively. Hence, we use `geom_line` with the aesthetic encoding given in `aes()` below. We manually matched the colors using `scale_color_manual`; we found these HEX color codes using an online image color picker. By default, the x-axis would show the year associated with each date, but these years are not meaningful in the current data – the lines are averages over three years. Therefore, we print only the month name by setting the `date_labels` argument within `scale_x_date()`.

```
temperature <- read_csv("https://raw.githubusercontent.com/krisrs1128/stat479_s22/main/data/temperature
ggplot(temperature) +
  geom_line(aes(date, temperature, col = city)) +
  scale_color_manual(values = c("#019e73", "#56b4e9", "#e7a208", "#cc79a7")) +
  scale_x_date(date_labels = "%b") +
  theme_minimal()
```



b. Using the `group_by` + `summarise` pattern, compute the mean temperature for each month in each city.

The `group_by` + `summarise` pattern is difficult to directly apply in this context, because we don't have a month variable to group by! However, if we look carefully, we can find that the month names are available through the `date` variable. Therefore, we first derive a month variable with which to use for grouping, before taking the average of all temperatures within that month.

```
averages <- temperature %>%
  group_by(new_month = month(date, label = TRUE), city) %>%
  summarise(mean_temp = mean(temperature))

head(averages, 4)
```

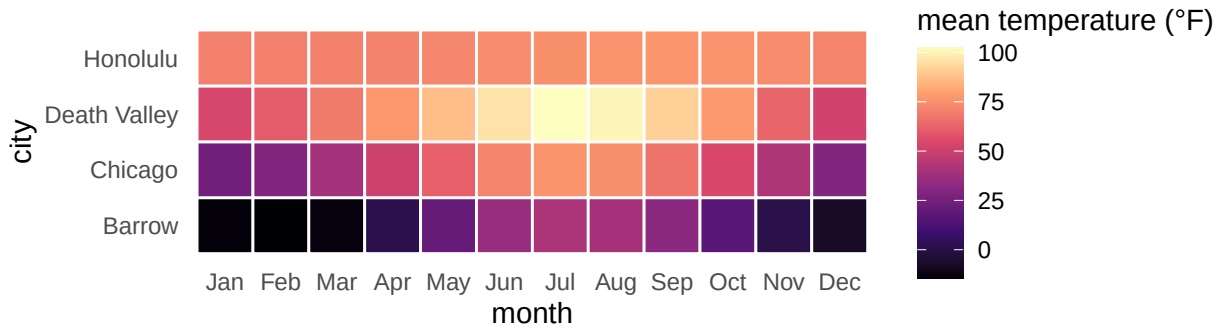
```
## # A tibble: 4 x 3
## # Groups:   new_month [1]
##   new_month city         mean_temp
##   <ord>     <chr>            <dbl>
## 1 Jan       Barrow           -13.4
## 2 Jan       Chicago           24.8
## 3 Jan       Death Valley      53.5
## 4 Jan       Honolulu          69.9
```

c. Using the data generated in (b), Make a version of Figure 2.4 using a tile mark (`geom_tile`). Try either (i) adding the `scale_fill_viridis_c(option = "magma")` scale to match the color scheme from the reading or (ii) adding `coord_fixed()` to make sure the marks are squares, not rectangles.

The `x`, `y`, and `color` channels encode month, city, and average temperature, respectively, and this is reflected in our `aes()`. We set `col = "white"` to create a thin white border around each of the squares. We make both changes (i) and (ii) exactly as discussed in the problem description. Finally, we remove superfluous tick and grid information using our `theme()` and add meaningful axes labels through `labs`.

```
ggplot(averages) +
  geom_tile(aes(new_month, city, fill = mean_temp), col = "white", lwd = 0.5) +
  scale_fill_viridis_c(option = "magma") +
  coord_fixed() +
  theme_minimal() +
```

```
  theme(axis.ticks = element_blank(), panel.grid = element_blank()) +
  labs(x = "month", fill = "mean temperature (°F)")
```



d. Compare and contrast the two displays. What types of comparisons are easier to make / what patterns are most readily visible using Figure 2.3 vs. Figure 2.4, and vice versa?

The line plot emphasizes change in temperature – our eyes more easily follow the rise and fall within the lines, compared to the darkening and lightning in color used in the heatmap. Differences in temperatures during specific dates are also more readily apparent, as are queries like "During which seasons is the temperature within a range of [10, 20] degrees?" Finally, only the line plot reveals any information at a sub-monthly granularity.

However, though the heatmap does not allow such precise comparisons, it is much more space efficient. In principle, we could have included many more cities and still have an interpretable plot. It is also possible to shrink this plot to take up a smaller space on the page while still maintaining readability, a potentially important consideration in case we have limited printed / screen space available within the larger report.

Soccer Code Review [3 points]

This exercise asks you to conduct an imaginary code review. These are often used in data science teams to,

- *Catch potential bugs*
- *Make sure code is transparent to others*
- *Create a shared knowledge base*

It is important to be perceptive but friendly.

- *Can the code be made more compact?*
- *Are there visual design choices / encodings that could be refined?*
- *If your colleague did something well, say so!*

They can also be a great way to learn new functions and programming patterns. Unlike standard code-reviews, I ask you to give an example implementing your recommendations.

Specifically, in this review, suppose you are working on a sports blog, and your colleague is soccer interested in which teams won the most games in a few European leagues over the last few years. They have written the code below. Provide your code review as a set of bullet points, and include code giving an example implementation of your ideas. The original data are from this link.

```
win_props <- read_csv("https://raw.githubusercontent.com/krisrs1128/stat479_s22/main/exercises/data/und
  group_by(team, year) %>%
  summarise(n_games = n(), wins = sum(wins) / n_games)

best_teams <- win_props %>%
  ungroup() %>%
```

```
  slice_max(wins, prop = 0.2) %>%
  pull(team)

win_props %>%
  filter(team %in% best_teams) %>%
  ggplot() +
  geom_point(aes(year, team, size = n_games, alpha = wins))
```

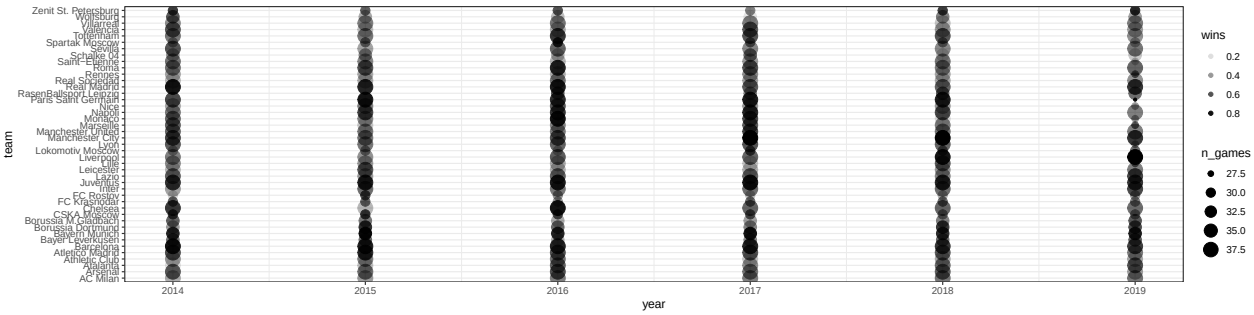
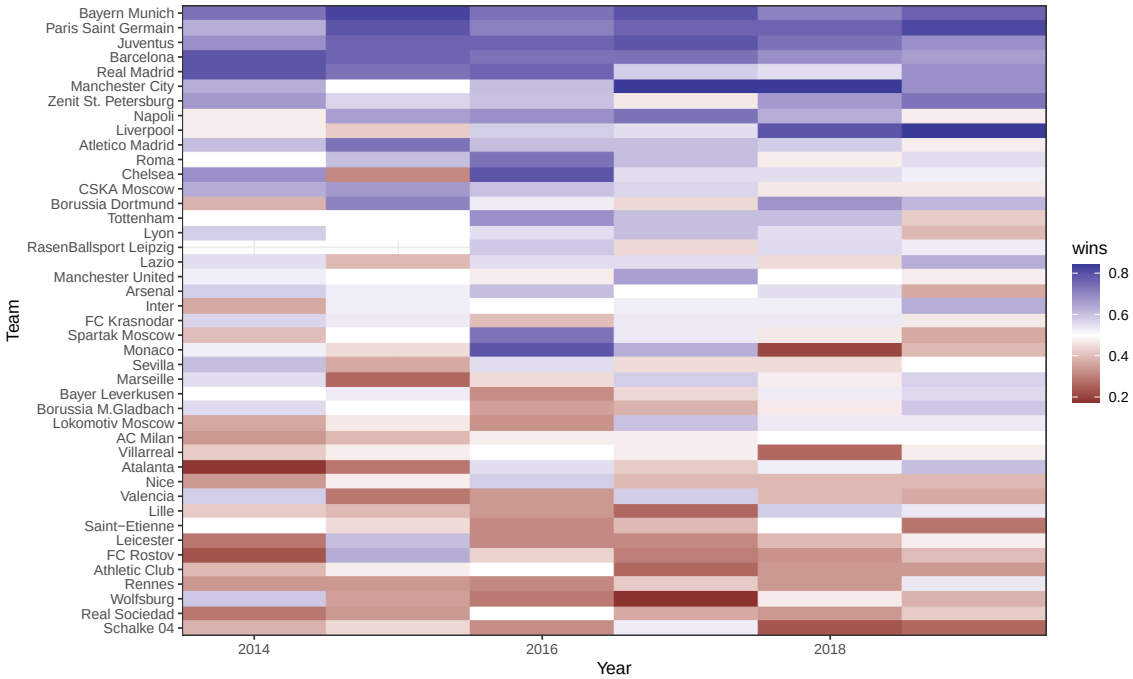


Figure 1: An example figure for code review.

Though the original code is reasonably consise, the visual design is far from ideal. The most important limitation is that it is very difficult to distinguish differences in point size and transparency, meaning that we can't easily answer which teams are actually the best. Second, the data only occupy a very small portion of the figure's available space – there are large gaps between observations – and overall, the design poorly uses the available visual real estate.

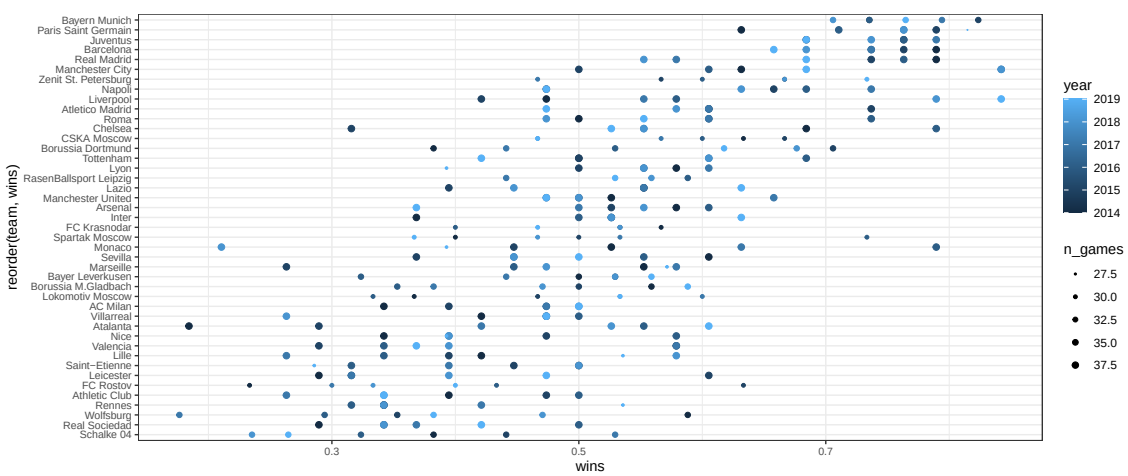
We propose two visual alternatives. The first uses a heatmap layout, but shows the same data. In comparison to transparency, differences in color are easier to recognize. Moreover, by sorting teams by their average number of wins, we can more easily navigate the ranking. The use of rectangular heatmap tiles instead of points also resolves the issue of empty visual space. Note that, in our implementation below, we manually set the color palette's midpoint to 0.5. Teams are in different colors depending on whether they win more games than they lose.

```
win_props <- filter(win_props, team %in% best_teams)
ggplot(win_props) +
  geom_tile(aes(year, reorder(team, wins), fill = wins, col = wins)) +
  scale_x_continuous(expand = c(0, 0)) +
  scale_y_discrete(expand = c(0, 0)) +
  scale_fill_gradient2(midpoint = 0.5) +
  scale_color_gradient2(midpoint = 0.5) +
  labs(x = "Year", y = "Team")
```



Our second approach gives up on encoding year positionally. This allows us to present the fraction of wins on the x-axis – arguably a more important variable of interest. This allows us to make more subtle distinctions in the fraction of wins than was possible when we were simply using color or transparency, as in the previous versions of the figure. Note the use of `scale_size` in the implementation below. This makes the points smaller than they would have been by default.

```
ggplot(win_props) +
  geom_point(aes(wins, reorder(team, wins), size = n_games, col = year)) +
  scale_size(range = c(0, 2)) +
  theme(axis.text = element_text(size = 8))
```



Visual Redesign [5 points]

Antibiotics Comparison [3 points]

Below, we provide three approaches to visualizing species abundance over time in an antibiotics dataset.

```r
antibiotic <- read_csv("https://uwmadison.box.com/shared/static/5jmd9pku62291ek20lioevsw1c588ahx.csv")
head(antibiotic, 4)
```

```
## # A tibble: 4 x 7
##   species  sample value ind    time svalue antibiotic
##   <chr>    <chr>  <dbl> <chr> <dbl>  <dbl> <chr>
## 1 Unc05qi6 D1         0 D         1     NA Antibiotic-free
## 2 Unc05qi6 D2         0 D         2     NA Antibiotic-free
## 3 Unc05qi6 D3         0 D         3      0 Antibiotic-free
## 4 Unc05qi6 D4         0 D         4      0 Antibiotic-free
```

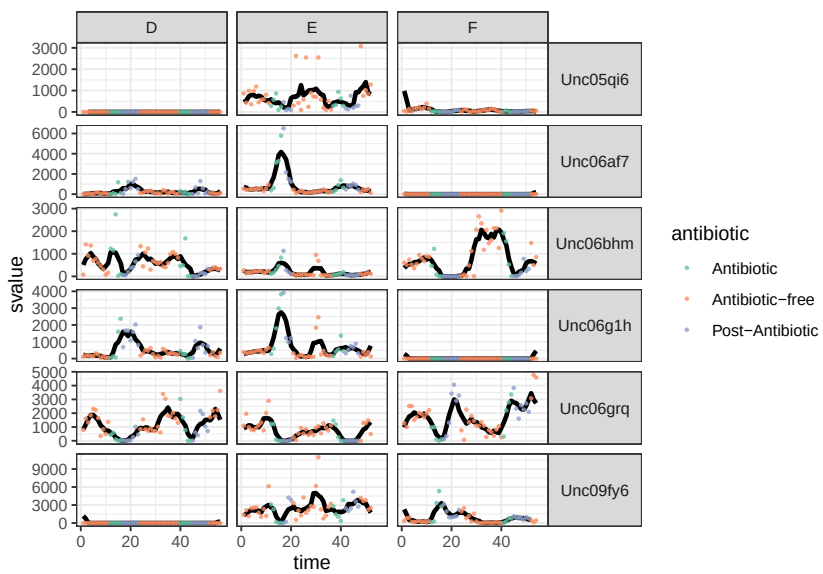
For each approach, describe,

** One type of visual comparison for which the visualization is well-suited. * One type of visual comparison for which the visualization is poorly-suited.*

Make sure to explain your reasoning.

a. [1.5 points] Approach 1

```r
ggplot(antibiotic, aes(x = time)) +
  geom_line(aes(y = svalue), size = 1.2) +
  geom_point(aes(y = value, col = antibiotic), size = 0.5, alpha = 0.8) +
  facet_grid(species ~ ind, scale = "free_y") +
  scale_color_brewer(palette = "Set2") +
  theme(strip.text.y = element_text(angle = 0))
```



Effective:

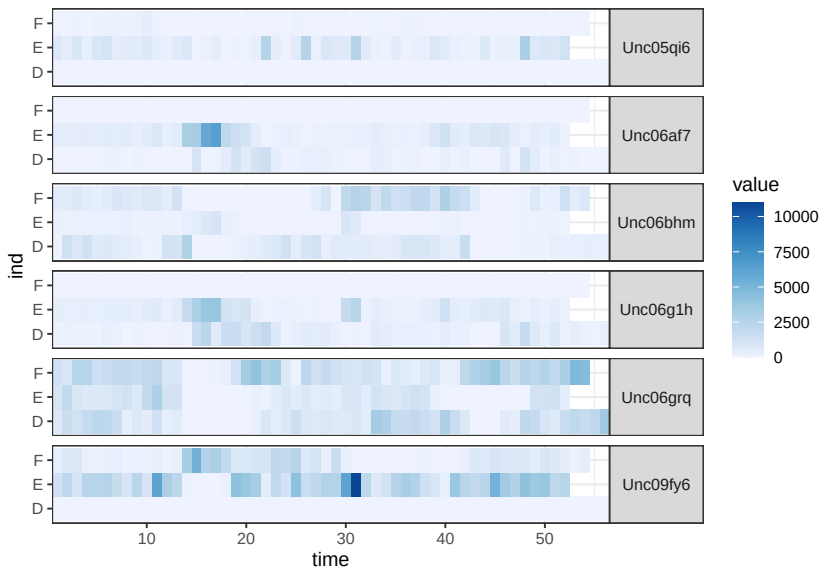
- Comparing abundances over time for each species and subject combination, even for rare species. It is easy to compare y-axis values within individual panels. Since the y-axis scales are not shared, trends in even the rare species are visible.
- Comparing species abundance across antibiotic treatment regimes. Since color is used to encode treatment regime, we can easily see how peaks or valleys coincide with the treatments.

Ineffective:

- Comparing abundances of different species for the same subject. Since the y-axes scales are not shared, it is hard to compare abundances across species.
- Ranking species by overall abundance within or across subjects. Again, this is a consequence of the unshared axis scales.
- Comparing trends in species abundances across subjects (especially D vs. F). Since our eyes have to travel left and right to compare species trends, it is harder to evaluate differences across subjects, compared to if they were all overlapping, for example.

b. [1.5 points] Approach 2

```
ggplot(antibiotic) +
  geom_tile(aes(time, ind, fill = value)) +
  scale_x_continuous(expand = c(0, 0)) +
  scale_fill_distiller(direction = 1) +
  facet_grid(species ~ .) +
  theme(strip.text.y = element_text(angle = 0))
```



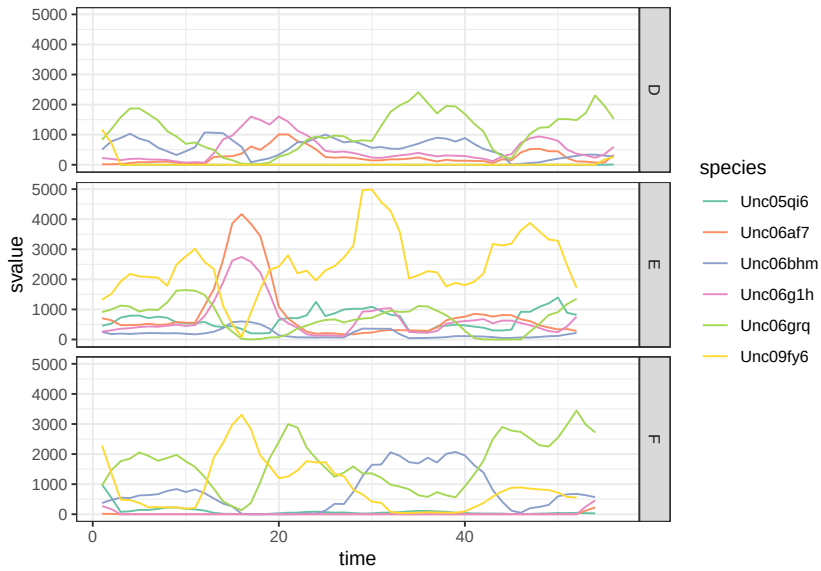
Effective

- For individual species, comparing trends over time across subjects. All the subjects are placed adjacent to one another within each panel, so our eyes don't have to travel such a large distance to make the comparison.
- Across species, recognizing shared increases or decreases at specific timepoints. Since the plot is so compact, all the values for a single timepoint are easily queryable.
- Recognizing the species and samples with the highest abundances. The cells with the darkest colors pop out from among the rest.

Ineffective

- Comparing the absolute abundances of a single species over time. It is difficult to compare shades of the same color.
- Evaluating the abundance of relatively rare species. These species all have light colors, and gradations smaller than the color scale bin size are not visible.
- Comparing species abundances for a single subject. We have to move our eyes across the three panels to make comparisons about a single species.

c. [1.5 points] Approach 3



Effective

- Within a single subject, ranking species by overall abundance. We can easily see which colors lie above the others within any given panel.
- Comparing abundance over time for a single subject and species. We can see increases and decreases clearly when plotting against a y-axis scale.
- Comparing overall species abundances across subjects. Since the same y-axis scale is used across panels, we can conclude that some subjects have more counts overall.

Ineffective

- Comparing trends for a single species across subjects. It is visually challenging to match colors across the three panels.
- Comparing trends for low abundance species. For low abundances, many of the lines overlap with one another.

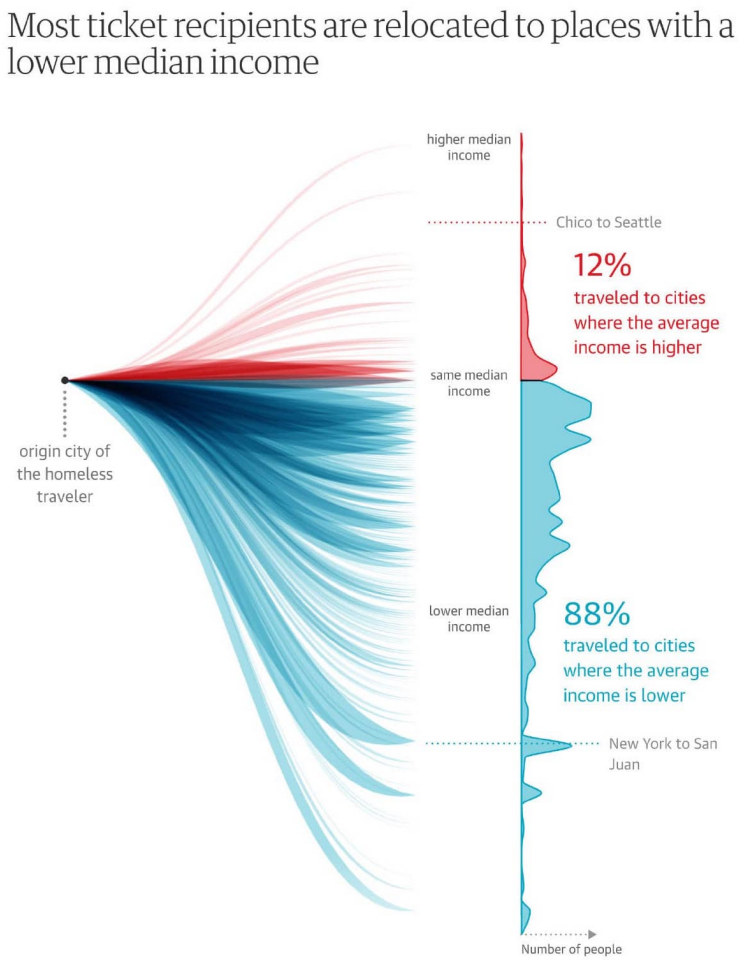
d. [1.5 points] Sketch code that could be used to make one of the three visualizations above.

The code is given in the blocks above.

Homelessness [3 points]

Take a static screenshot from any of the visualizations in this article, and deconstruct its associated visual encodings.

We will analyze this screenshot,



a. What do you think was the underlying data behind the current view? What where the rows, and what were the columns?

The data would have had to include origin-destination city pairs for all the individuals who were bussed out, the median incomes in each of those cities, and either (i) the number of people associated with each origin-destination pair or (ii) a person-level record of trips. These data could have been formed from a join between two tables: one including all origin-destination counts and another including city names with their median incomes. In the first of these tables, the rows would be specific travel routes, the columns might have been `origin`, `destination` and `count`. In the second table, each row is a city, and the columns might have been `city` and `median_income`.

b. What were the data types of each of the columns?

- `origin`: Categorical
- `destination`: Categorical
- `count`: Numeric (positive integer)
- `median_income`: Numeric (positive real)

c. What encodings were used? How are properties of marks on the page derived from the underlying abstract data?

- Edge x-positions: Distinguishes between origin and destination of travel.

- Edge and histogram y-positions: What is the difference in median income between the origin and destination city?
- Edge thickness: How many of the records followed that edge's origin-destination pair?
- Histogram x-position: The distance from the base of the histogram describes how many cities had that particular difference between origin an ddestination incomes.
- Color: Does the destination city have a higher median income than the source city?

In addition to these purely graphical elements, there is annotation that describes both how to read the figure ("origin city of the homeless traveler") and its main takeaways (12% vs. 88% higher vs. lower income destinations).

d. Is multi-view composition being used? If so, how?

Yes, this is a compound figure. The left hand side is a weighted network visualization while the right hand side is a histogram. The two figures share the same color and y-axis encodings, as described in the previous part. This makes it possible to analyze specific origin-destination pairs together with an overall summary of changes. This design can be justified both aesthetically – the edges in the network evoke the mapped routes shown earlier in the piece – as well as formally, as a way of giving route-level details (the network) together with population-level overviews (the histogram).